SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 4 – Experiments (20 Questions)

Course Code:	ITA0332	**Course Title:**	Mobile computing
CO Assessed:	CO2 – Manipulation (BL4)	**Assessment:**	Assessment Tool 4 – Experiments
Weightage:	40 % of CIA	**Total Marks:**	20 (Answer ANY ONE) 1 x 20 = 20
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications. (BTL 4)*		
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Section / Batch:	Slot-A/B.E-AIDS-2024-28	**Date:**	3/11/2026

Code of Conduct

I CH.V.Ramayogi/192424319 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ch.V.Ramayogi

Instructions

- This test contains 20 questions (ANSWER ANY ONE), Total: 20 Marks.
- No calculators or electronic devices permitted.
- Duration: 30 minutes.

QUESTION 1: Experiment based on Multiple Access (FDMA/TDMA/CDMA/OFDMA)

1. Aim

To compare FDMA, TDMA, CDMA and OFDMA when multiple users share a wireless channel.

2. Theory / Concept

FDMA assigns a separate frequency band to each user; guard bands reduce adjacent-channel interference.

TDMA assigns users different time slots on a shared carrier; synchronization is essential.

CDMA allows users to transmit over the same band using different spreading codes; code orthogonality and power control are important.

OFDMA divides the channel into orthogonal subcarriers and allocates groups of subcarriers to users. It is flexible for varying traffic.

3. Tools and Experimental Setup

Use a wireless-system simulator (for example, MATLAB/Simulink, ns-3 or an equivalent lab simulator). Configure four users, a common channel bandwidth, transmit power, receiver noise, path loss and an interference level. Keep the traffic load and channel conditions identical for a fair comparison.

4. Procedure

Create four user nodes and configure the same offered traffic for each user.

Run the simulation separately using FDMA, TDMA, CDMA and OFDMA.

Record bandwidth allocation, received signal strength, throughput, average delay, bit error rate (BER) and spectral efficiency.

Increase the number of users and offered load in stages; repeat the measurements.

Increase noise/interference and then introduce mobility or changing path loss. Record the effect on each technique.

Plot throughput, delay and BER against user count or signal-to-noise ratio (SNR), and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
FDMA	Fixed frequency allocation; guard bands consume some bandwidth; performance can be predictable at low load.
TDMA	Time-slot sharing; delay can rise when users wait for their assigned slots; timing synchronization is

	required.
CDMA	Users share time and frequency; interference rises as users increase; power control helps limit the near-far problem.
OFDMA	Subcarriers can be allocated dynamically; generally supports efficient scheduling and high data rates in broadband systems.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As the number of active users and interference increase, shared resources become more heavily loaded. FDMA may waste capacity when a user is idle; TDMA can add waiting time; CDMA can become interference-limited; OFDMA can adapt resource blocks to traffic but requires accurate synchronization and channel estimation. Higher noise generally increases BER and retransmissions, reducing useful throughput.

7. Optimization Techniques

Use adaptive modulation and coding, efficient scheduling, power control, channel estimation, interference coordination and suitable guard intervals. In a mobile environment, use handover and channel tracking where supported.

8. Result / Conclusion

The experiment demonstrates that no single multiple-access method is best for every condition. OFDMA offers flexible resource allocation for broadband traffic, while FDMA, TDMA and CDMA illustrate frequency-, time- and code-based sharing. The best choice depends on user density, mobility, interference, delay and bandwidth requirements.

QUESTION 2: Experiment based on Radio Interfaces

1. Aim

To study a cellular radio interface and evaluate how its radio parameters affect communication performance.

2. Theory / Concept

A radio interface defines how a mobile device communicates over the air with a base station, including the physical channel, modulation, coding, access method and radio-resource procedures.

Important configurable parameters include carrier frequency, channel bandwidth, transmit power, modulation, coding rate, noise and interference.

The received signal quality is affected by path loss, fading, shadowing, obstacles and user mobility.

3. Tools and Experimental Setup

Use a cellular radio-interface simulator or a link-level/network simulator. Configure a mobile station, base station, carrier frequency, channel bandwidth, transmit power, receiver noise and a traffic source. Select a supported modulation and coding scheme.

4. Procedure

Establish a baseline link with one stationary user and record received signal strength and SNR.

Generate data traffic and measure throughput, delay and BER.

Change bandwidth while keeping other settings constant; record the change in throughput and spectral efficiency.

Vary transmit power and receiver noise separately, noting the effect on SNR and BER.

Add a second and then additional users to observe resource sharing and interference.

Introduce mobility and obstacles/fading in the simulator; observe signal variation and any handover or link adaptation.

5. Observation

Parameter / Feature	**Expected observation**
Bandwidth	A wider channel can support a higher peak data rate, subject to radio conditions and system limits.
Signal strength / SNR	A stronger desired signal relative to noise generally improves decoding and lowers BER.
Interference	Co-channel or adjacent-channel interference can reduce SINR, throughput and reliability.
Mobility	Movement changes path loss and fading; link

	adaptation or handover may be needed.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput is the successfully delivered data per unit time. Average delay is the time taken for packets to reach the destination. BER is erroneous bits divided by total transmitted bits. Spectral efficiency is throughput per unit bandwidth, commonly expressed in bit/s/Hz. The results should show that radio quality, bandwidth, coding and user load jointly determine performance.

7. Optimization Techniques

Apply power control, adaptive modulation/coding, robust channel coding, interference coordination, antenna diversity/beamforming where available, and appropriate handover thresholds.

8. Result / Conclusion

The radio-interface experiment shows how physical-layer settings and radio conditions affect throughput, delay, BER and spectral efficiency. Reliable performance requires balancing data rate, coverage, interference and mobility.

QUESTION 3: Experiment based on Protocols

1. Aim

To analyze the role of mobile-network protocols in reliable data communication and measure their performance under changing wireless conditions.

2. Theory / Concept

Protocols define the rules for addressing, packet formatting, transmission, error handling, routing and delivery.

A mobile data path commonly involves application, transport, network and link/physical functions.

TCP provides reliable, ordered delivery using acknowledgements and retransmissions; UDP has lower protocol overhead but does not itself guarantee delivery.

Wireless losses, variable delay and mobility can affect protocol performance.

3. Tools and Experimental Setup

Use a network simulator such as ns-3 or an equivalent tool. Create a mobile node, an access point/base station, a server and a traffic application. Configure a wireless link, IP addressing and either TCP or UDP traffic.

4. Procedure

Build the topology and verify that the mobile node can reach the server.

Run a baseline TCP flow and record throughput, packet delivery, delay and packet loss.

Repeat using UDP with the same offered traffic rate.

Increase background traffic and the number of mobile users; record congestion effects.

Add wireless errors, higher delay and mobility; observe retransmissions, packet loss and throughput.

Compare TCP and UDP results and explain the effect of reliability mechanisms.

5. Observation

Parameter / Feature	**Expected observation**
TCP	Acknowledgements and retransmissions improve reliable delivery, but can increase delay and reduce throughput on lossy links.
UDP	Lower overhead and no built-in retransmission; delay may be lower, but the application must tolerate or handle loss.
Congestion	Higher offered load can cause queueing delay, packet drops and reduced goodput.

Mobility	Route changes or temporary disconnections can interrupt flows and trigger recovery.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput/goodput measures useful delivered data. Delay includes transmission, propagation, processing and queueing time. Packet loss and retransmissions become more significant under congestion or weak radio conditions. TCP is suitable where complete ordered delivery is important; UDP is useful for delay-sensitive real-time traffic when the application can manage losses.

7. Optimization Techniques

Use suitable transport protocols, congestion control, traffic prioritization, buffering, robust wireless links, mobility-aware routing and application-level error handling where needed.

8. Result / Conclusion

The experiment highlights that protocol selection affects reliability, delay and throughput. TCP prioritizes reliable ordered delivery, while UDP offers lower overhead and is often selected for real-time applications with suitable loss handling.

QUESTION 4: Experiment based on GPRS

1. Aim

To study General Packet Radio Service (GPRS), a packet-switched data service introduced for GSM networks, and evaluate its performance.

2. Theory / Concept

GPRS adds packet-data capability to GSM and allows radio resources to be shared by users when data is sent.

Its network architecture includes the Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN), which support mobility/session management and connection to external packet networks.

GPRS uses allocated GSM timeslots and coding schemes; actual data rate depends on radio conditions, timeslot allocation and network load.

3. Tools and Experimental Setup

Use a GSM/GPRS network simulator or a suitable teaching simulation. Model a mobile station, BTS/BSC, SGSN, GGSN and packet-data server. Configure available timeslots, coding scheme, signal quality, traffic load and background users.

4. Procedure

Create a packet-data session between the mobile station and server through the GPRS core.

Run a small data transfer and record throughput, packet delay and packet loss/BER if available.

Increase the number of active users and vary the timeslots allocated to the test user.

Change signal quality or coding scheme and repeat the transfer.

Introduce movement between cells and observe session continuity and mobility handling.

Compare lightly loaded and heavily loaded network results.

5. Observation

Parameter / Feature	**Expected observation**
Packet switching	Radio capacity is used when packets are sent, allowing statistical sharing among users.
Timeslots	More available timeslots can increase the user's data capacity, subject to network configuration.
Radio quality	Poor signal quality can require more robust coding and reduce effective data rate.
Network load	More simultaneous users can increase waiting

	time and reduce per-user throughput.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

GPRS is designed for packet data rather than a permanently dedicated circuit. Its throughput is modest compared with later 3G/4G/5G systems and varies with timeslot allocation, coding, signal quality and congestion. Mobility procedures help maintain packet service as a user changes cells, although interruptions may occur.

7. Optimization Techniques

Allocate timeslots according to demand, improve radio coverage, manage congestion, select appropriate coding schemes and maintain efficient mobility/session management.

8. Result / Conclusion

The experiment demonstrates packet-switched data delivery over GSM/GPRS. Performance depends on timeslot availability, coding scheme, radio quality and the number of active users.

QUESTION 5: Experiment based on HSCSD

1. Aim

To study High-Speed Circuit-Switched Data (HSCSD) and evaluate how combining GSM timeslots affects data performance.

2. Theory / Concept

HSCSD is a GSM enhancement that increases circuit-switched data capacity by assigning multiple timeslots to one connection.

Unlike packet-switched GPRS, HSCSD reserves circuit resources for the duration of the connection.

The achievable data rate depends on the number of allocated timeslots and the channel coding used.

3. Tools and Experimental Setup

Use a GSM/HSCSD teaching simulator. Configure a mobile station and base station, a circuit-switched data connection, available timeslots, coding rate, signal quality and competing users.

4. Procedure

Establish a baseline data connection using one timeslot and record throughput and delay.

Increase the allocated timeslots step by step, within the simulator's supported limit.

Measure the resulting data rate and resource usage for each allocation.

Add competing users and observe whether sufficient circuit resources remain available.

Reduce signal quality or add interference and repeat the measurements.

Compare HSCSD's reserved-resource behavior with packet-sharing behavior conceptually.

5. Observation

Parameter / Feature	**Expected observation**
Timeslot allocation	Allocating more timeslots increases the potential data rate for the connection.
Resource reservation	Reserved circuit resources may remain unavailable to other users even during idle periods.
Interference	Poor radio conditions can reduce effective performance or increase errors.
User capacity	A connection using several timeslots consumes more network resources.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

HSCSD can provide a higher circuit-switched data rate than a basic GSM data connection by combining timeslots. Its limitation is inefficient resource use for bursty data because resources are reserved for the session. The number of available timeslots and radio quality constrain performance.

7. Optimization Techniques

Choose the number of timeslots according to application demand, maintain good radio coverage and manage circuit-resource allocation to avoid excessive blocking.

8. Result / Conclusion

The experiment verifies that HSCSD increases circuit-switched data capacity through multi-timeslot allocation, but it uses dedicated resources and is less flexible for intermittent packet traffic than packet-switched services.

QUESTION 6: Experiment based on 1G

1. Aim

To examine first-generation (1G) mobile communication and evaluate its basic analog voice characteristics.

2. Theory / Concept

1G systems were the first generation of cellular mobile networks and used analog radio transmission mainly for voice.

Common 1G systems included AMPS, NMT and TACS in different regions.

Frequency Division Multiple Access (FDMA) was commonly used to assign separate frequency channels to calls.

1G had limited capacity, weak security and lower voice quality compared with digital generations.

3. Tools and Experimental Setup

Use a cellular-system simulator or a comparative model of an analog FDMA network. Configure a cell, frequency channels, transmit power, noise, path loss and multiple mobile users.

4. Procedure

Create a cell and assign a separate frequency channel to each active call.

Start one voice connection and record signal strength and voice-link quality.

Add users until the available channels are occupied; record blocking or call rejection.

Increase noise and co-channel interference and observe voice quality.

Move a mobile user toward the cell edge and observe signal degradation and handover behavior if modeled.

Compare the system's capacity, security and service type with later digital generations.

5. Observation

Parameter / Feature	**Expected observation**
Access method	FDMA assigns separate frequency channels to calls.
Service	Primarily analog voice; data services were very limited.
Capacity	Limited by the number of available radio channels.
Security and quality	Analog transmission offered weaker privacy and was more susceptible to noise.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As users increase, the finite number of frequency channels limits capacity. Noise and interference directly affect analog voice quality. 1G demonstrates the basic cellular concept but lacks the spectral efficiency, digital security and data capabilities of later generations.

7. Optimization Techniques

Within a legacy 1G model, careful frequency planning, suitable channel reuse and adequate coverage can reduce interference. The major long-term improvement is migration to digital cellular technologies.

8. Result / Conclusion

The experiment shows that 1G enabled mobile analog voice communication using cellular frequency channels, but its capacity, security and service capabilities were limited.

QUESTION 7: Experiment based on 2G

1. Aim

To study second-generation (2G) digital cellular communication and compare its performance with analog 1G systems.

2. Theory / Concept

2G introduced digital voice transmission and improved capacity, security and service quality.

GSM commonly uses TDMA with FDMA carrier allocation; IS-95 uses CDMA.

Digital systems support services such as SMS and, with enhancements, packet data.

Speech coding, channel coding and encryption improve efficiency and communication security.

3. Tools and Experimental Setup

Use a GSM/2G or equivalent cellular simulator. Configure a base station, mobile users, carrier channels, time slots, signal strength, noise and traffic. Select voice or SMS/data traffic.

4. Procedure

Configure a digital 2G cell and register a mobile user.

Establish a voice call or send a test message; verify successful delivery.

Add users and observe channel/time-slot allocation and call blocking.

Vary signal strength and interference; record BER, call quality and throughput for data traffic.

Introduce mobility and observe handover between cells.

Compare digital 2G performance with the analog 1G baseline.

5. Observation

Parameter / Feature	**Expected observation**
Digital transmission	Voice is represented digitally, allowing coding and error protection.
Capacity	TDMA/FDMA or CDMA methods support more organized sharing than basic analog systems.
Services	Voice and SMS are supported; packet-data enhancements include GPRS and EDGE.
Security	Digital authentication and encryption provide better protection than 1G.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Digital coding can improve voice quality when radio errors are within the system's correction capability. Under weak signal conditions, BER rises and service quality can fall. Capacity depends on the access method, channel allocation, interference and network configuration.

7. Optimization Techniques

Use proper frequency planning, power control, channel coding, interference management and efficient handover configuration.

8. Result / Conclusion

The experiment demonstrates that 2G improved mobile communication through digital transmission, better security, increased capacity and services such as SMS, with later enhancements supporting packet data.

QUESTION 8: Experiment based on 3G

1. Aim

To study third-generation (3G) mobile networks and evaluate their support for voice and mobile data.

2. Theory / Concept

3G networks were designed to provide higher data rates and multimedia services in addition to voice.

UMTS commonly uses WCDMA radio access; CDMA2000 is another 3G family.

3G supports services such as mobile internet, video calling and multimedia messaging.

Packet data and improved radio techniques increase capability compared with 2G.

3. Tools and Experimental Setup

Use a UMTS/WCDMA or 3G network simulator. Model a user equipment device, Node B, radio network controller/core network and a data server. Configure channel bandwidth, transmit power, noise, interference, mobility and traffic.

4. Procedure

Register a mobile device with the 3G network and establish a data session.

Run a web/data transfer and record throughput, delay and packet loss or BER.

Add multiple users and vary their traffic load.

Change signal strength, interference and mobility; repeat the measurements.

Test a voice or multimedia session if the simulator supports it.

Compare 3G results with 2G, keeping traffic and test conditions as similar as possible.

5. Observation

Parameter / Feature	**Expected observation**
Radio access	WCDMA spreads users over a wider carrier and separates them using codes.
Data service	Supports mobile internet and multimedia applications at higher rates than typical 2G services.
Interference	Cell loading and interference affect the shared WCDMA capacity.
Mobility	Handover and radio-resource control support users moving between cells.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

3G generally provides higher data capacity than 2G, but actual throughput depends on radio conditions, cell load, user mobility and service configuration. Increased interference and user count can reduce data rates and increase delay.

7. Optimization Techniques

Use power control, admission control, suitable spreading codes, load balancing, robust coding and well-planned handover parameters.

8. Result / Conclusion

The experiment shows that 3G extends mobile networks beyond basic voice and messaging by enabling practical mobile internet and multimedia services, with performance dependent on radio quality and cell loading.

QUESTION 9: Experiment based on 4G

1. Aim

To study fourth-generation (4G LTE) mobile communication and measure broadband data performance.

2. Theory / Concept

4G LTE is an all-IP mobile broadband system designed for high data rates, lower latency and efficient packet communication.

LTE uses OFDMA in the downlink and SC-FDMA in the uplink.

MIMO, adaptive modulation and coding, and flexible bandwidths improve spectral efficiency and capacity.

Voice can be provided using Voice over LTE (VoLTE) where deployed.

3. Tools and Experimental Setup

Use an LTE network simulator. Configure user equipment, eNodeB, EPC/core network and a server. Set channel bandwidth, resource blocks, modulation/coding, transmit power, noise, interference and mobility.

4. Procedure

Attach the user equipment to the LTE network and establish an IP data connection.

Run a data transfer and record throughput, average delay, packet loss and BER if available.

Increase the number of users and offered traffic; observe scheduling and resource sharing.

Change bandwidth, signal quality and interference one at a time.

Enable mobility and observe handover and data-session continuity.

Compare LTE results with 3G under equivalent traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
Downlink/uplink	LTE uses OFDMA downlink and SC-FDMA uplink.
Throughput	Broadband data rates are higher than typical 3G performance under suitable conditions.
Latency	An all-IP architecture and optimized radio procedures support lower latency.
Resource allocation	The scheduler assigns resource blocks according to channel quality and traffic demand.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

LTE performance improves with available bandwidth, good SINR, efficient scheduling and MIMO support. More users and interference compete for finite resources, which can lower per-user throughput and increase delay. Mobility requires handover between eNodeBs.

7. Optimization Techniques

Use channel-aware scheduling, MIMO, adaptive modulation/coding, interference coordination, load balancing and appropriate handover settings.

8. Result / Conclusion

The experiment demonstrates that 4G LTE provides high-speed IP-based mobile broadband with efficient OFDMA-based resource allocation and lower latency than earlier generations.

QUESTION 10: Experiment based on 5G

1. Aim

To study fifth-generation (5G) mobile communication and evaluate its high-speed, low-latency and massive-connectivity capabilities.

2. Theory / Concept

5G is designed to support enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC).

5G NR uses flexible OFDM-based radio technology, scalable numerology and advanced scheduling.

Massive MIMO and beamforming improve capacity and spatial coverage.

5G can operate in sub-6 GHz and millimeter-wave frequency ranges; higher frequencies can provide wide bandwidth but are more sensitive to blockage and propagation loss.

3. Tools and Experimental Setup

Use a 5G NR simulator or network simulation environment. Configure a gNodeB, user equipment, 5G core, carrier frequency, channel bandwidth, numerology, antenna configuration, traffic type, noise, interference and mobility.

4. Procedure

Configure a 5G cell and connect one user equipment device.

Run an eMBB data flow and record throughput, delay and packet loss/BER if supported.

Add users or IoT devices and observe resource allocation and per-user performance.

Test a low-latency traffic profile and measure end-to-end delay.

Vary frequency, bandwidth, beamforming, blockage and mobility; record the effect.

Compare the results with a 4G LTE baseline using comparable traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
eMBB	Supports high data rates for applications such as high-resolution streaming.
URLLC	Targets very low latency and high reliability for demanding control applications.
mMTC	Supports large numbers of connected devices with small, intermittent data transmissions.
Beamforming / MIMO	Can improve received signal quality and spatial reuse, depending on channel conditions.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

5G can achieve high throughput when wide bandwidth, good signal quality and suitable radio resources are available. Millimeter-wave links can be affected by blockage and short coverage range. Low latency depends on radio configuration, network architecture, load and application path; it is not guaranteed in every deployment.

7. Optimization Techniques

Use beam management, massive MIMO, suitable spectrum, edge computing for latency-sensitive services, QoS scheduling, interference coordination and dense small-cell deployment where justified.

8. Result / Conclusion

The experiment illustrates 5G's flexible support for broadband, low-latency and massive-IoT use cases. Actual performance depends on spectrum, deployment, signal conditions, network load and device capability.

QUESTION 11: Experiment based on Spread Spectrum (DSSS/FHSS)

1. Aim

To compare Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) under noise and interference.

2. Theory / Concept

Spread spectrum transmits a signal over a bandwidth wider than the minimum required for the information.

DSSS multiplies the data by a high-rate spreading code; the receiver uses the same code to despread the signal.

FHSS changes the carrier frequency according to a predefined hopping sequence known to transmitter and receiver.

Both methods can provide resistance to certain forms of interference, but their performance depends on codes, hopping pattern, synchronization and channel conditions.

3. Tools and Experimental Setup

Use a digital communication simulator. Configure a bit source, DSSS and FHSS transmitters/receivers, spreading code or hopping sequence, channel bandwidth, transmit power, AWGN noise and an interference source. Use the same data and comparable power for both tests.

4. Procedure

Generate a known binary data sequence and transmit it using DSSS.

At the receiver, synchronize the spreading code and despread the received signal.

Repeat using FHSS with a shared hopping sequence and synchronized transmitter/receiver.

Measure BER, throughput and occupied bandwidth with no added interference.

Add narrowband interference for one run and broadband noise for another; repeat the measurements.

Change the spreading factor or hopping rate and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
DSSS	Spreads each data symbol using a code; processing gain can reduce the effect of narrowband interference.
FHSS	Moves the carrier among frequencies; narrowband interference may affect only some hops.

Synchronization	Code or hopping-sequence mismatch prevents reliable recovery.
Bandwidth	Both methods use more spectrum than the unspread information signal.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

A higher spreading factor can improve processing gain but uses more chips per information bit and may reduce the net data rate for a fixed chip rate. FHSS performance depends on hop timing, number of channels and how often a hop encounters interference. Noise generally increases BER in both systems.

7. Optimization Techniques

Choose suitable spreading codes or hopping patterns, maintain synchronization, use appropriate transmit power and channel planning, and select spreading/hopping parameters according to interference conditions.

8. Result / Conclusion

The experiment shows that DSSS uses code-based spreading while FHSS uses frequency changes. Both can improve robustness against selected interference types, but neither eliminates all noise or interference.

QUESTION 12: Experiment based on Multiple Access (FDMA/TDMA/CDMA/OFDMA)

1. Aim

To compare FDMA, TDMA, CDMA and OFDMA when multiple users share a wireless channel.

2. Theory / Concept

FDMA assigns a separate frequency band to each user; guard bands reduce adjacent-channel interference.

TDMA assigns users different time slots on a shared carrier; synchronization is essential.

CDMA allows users to transmit over the same band using different spreading codes; code orthogonality and power control are important.

OFDMA divides the channel into orthogonal subcarriers and allocates groups of subcarriers to users. It is flexible for varying traffic.

3. Tools and Experimental Setup

Use a wireless-system simulator (for example, MATLAB/Simulink, ns-3 or an equivalent lab simulator). Configure four users, a common channel bandwidth, transmit power, receiver noise, path loss and an interference level. Keep the traffic load and channel conditions identical for a fair comparison.

4. Procedure

Create four user nodes and configure the same offered traffic for each user.

Run the simulation separately using FDMA, TDMA, CDMA and OFDMA.

Record bandwidth allocation, received signal strength, throughput, average delay, bit error rate (BER) and spectral efficiency.

Increase the number of users and offered load in stages; repeat the measurements.

Increase noise/interference and then introduce mobility or changing path loss. Record the effect on each technique.

Plot throughput, delay and BER against user count or signal-to-noise ratio (SNR), and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
FDMA	Fixed frequency allocation; guard bands consume some bandwidth; performance can be predictable at low load.
TDMA	Time-slot sharing; delay can rise when users wait for their assigned slots; timing synchronization is

	required.
CDMA	Users share time and frequency; interference rises as users increase; power control helps limit the near-far problem.
OFDMA	Subcarriers can be allocated dynamically; generally supports efficient scheduling and high data rates in broadband systems.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As the number of active users and interference increase, shared resources become more heavily loaded. FDMA may waste capacity when a user is idle; TDMA can add waiting time; CDMA can become interference-limited; OFDMA can adapt resource blocks to traffic but requires accurate synchronization and channel estimation. Higher noise generally increases BER and retransmissions, reducing useful throughput.

7. Optimization Techniques

Use adaptive modulation and coding, efficient scheduling, power control, channel estimation, interference coordination and suitable guard intervals. In a mobile environment, use handover and channel tracking where supported.

8. Result / Conclusion

The experiment demonstrates that no single multiple-access method is best for every condition. OFDMA offers flexible resource allocation for broadband traffic, while FDMA, TDMA and CDMA illustrate frequency-, time- and code-based sharing. The best choice depends on user density, mobility, interference, delay and bandwidth requirements.

QUESTION 13: Experiment based on Radio Interfaces

1. Aim

To study a cellular radio interface and evaluate how its radio parameters affect communication performance.

2. Theory / Concept

A radio interface defines how a mobile device communicates over the air with a base station, including the physical channel, modulation, coding, access method and radio-resource procedures.

Important configurable parameters include carrier frequency, channel bandwidth, transmit power, modulation, coding rate, noise and interference.

The received signal quality is affected by path loss, fading, shadowing, obstacles and user mobility.

3. Tools and Experimental Setup

Use a cellular radio-interface simulator or a link-level/network simulator. Configure a mobile station, base station, carrier frequency, channel bandwidth, transmit power, receiver noise and a traffic source. Select a supported modulation and coding scheme.

4. Procedure

Establish a baseline link with one stationary user and record received signal strength and SNR.

Generate data traffic and measure throughput, delay and BER.

Change bandwidth while keeping other settings constant; record the change in throughput and spectral efficiency.

Vary transmit power and receiver noise separately, noting the effect on SNR and BER.

Add a second and then additional users to observe resource sharing and interference.

Introduce mobility and obstacles/fading in the simulator; observe signal variation and any handover or link adaptation.

5. Observation

Parameter / Feature	**Expected observation**
Bandwidth	A wider channel can support a higher peak data rate, subject to radio conditions and system limits.
Signal strength / SNR	A stronger desired signal relative to noise generally improves decoding and lowers BER.
Interference	Co-channel or adjacent-channel interference can reduce SINR, throughput and reliability.
Mobility	Movement changes path loss and fading; link

	adaptation or handover may be needed.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput is the successfully delivered data per unit time. Average delay is the time taken for packets to reach the destination. BER is erroneous bits divided by total transmitted bits. Spectral efficiency is throughput per unit bandwidth, commonly expressed in bit/s/Hz. The results should show that radio quality, bandwidth, coding and user load jointly determine performance.

7. Optimization Techniques

Apply power control, adaptive modulation/coding, robust channel coding, interference coordination, antenna diversity/beamforming where available, and appropriate handover thresholds.

8. Result / Conclusion

The radio-interface experiment shows how physical-layer settings and radio conditions affect throughput, delay, BER and spectral efficiency. Reliable performance requires balancing data rate, coverage, interference and mobility.

QUESTION 14: Experiment based on Protocols

1. Aim

To analyze the role of mobile-network protocols in reliable data communication and measure their performance under changing wireless conditions.

2. Theory / Concept

Protocols define the rules for addressing, packet formatting, transmission, error handling, routing and delivery.

A mobile data path commonly involves application, transport, network and link/physical functions.

TCP provides reliable, ordered delivery using acknowledgements and retransmissions; UDP has lower protocol overhead but does not itself guarantee delivery.

Wireless losses, variable delay and mobility can affect protocol performance.

3. Tools and Experimental Setup

Use a network simulator such as ns-3 or an equivalent tool. Create a mobile node, an access point/base station, a server and a traffic application. Configure a wireless link, IP addressing and either TCP or UDP traffic.

4. Procedure

Build the topology and verify that the mobile node can reach the server.

Run a baseline TCP flow and record throughput, packet delivery, delay and packet loss.

Repeat using UDP with the same offered traffic rate.

Increase background traffic and the number of mobile users; record congestion effects.

Add wireless errors, higher delay and mobility; observe retransmissions, packet loss and throughput.

Compare TCP and UDP results and explain the effect of reliability mechanisms.

5. Observation

Parameter / Feature	**Expected observation**
TCP	Acknowledgements and retransmissions improve reliable delivery, but can increase delay and reduce throughput on lossy links.
UDP	Lower overhead and no built-in retransmission; delay may be lower, but the application must tolerate or handle loss.
Congestion	Higher offered load can cause queueing delay, packet drops and reduced goodput.

Mobility	Route changes or temporary disconnections can interrupt flows and trigger recovery.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput/goodput measures useful delivered data. Delay includes transmission, propagation, processing and queueing time. Packet loss and retransmissions become more significant under congestion or weak radio conditions. TCP is suitable where complete ordered delivery is important; UDP is useful for delay-sensitive real-time traffic when the application can manage losses.

7. Optimization Techniques

Use suitable transport protocols, congestion control, traffic prioritization, buffering, robust wireless links, mobility-aware routing and application-level error handling where needed.

8. Result / Conclusion

The experiment highlights that protocol selection affects reliability, delay and throughput. TCP prioritizes reliable ordered delivery, while UDP offers lower overhead and is often selected for real-time applications with suitable loss handling.

QUESTION 15: Experiment based on GPRS

1. Aim

To study General Packet Radio Service (GPRS), a packet-switched data service introduced for GSM networks, and evaluate its performance.

2. Theory / Concept

GPRS adds packet-data capability to GSM and allows radio resources to be shared by users when data is sent.

Its network architecture includes the Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN), which support mobility/session management and connection to external packet networks.

GPRS uses allocated GSM timeslots and coding schemes; actual data rate depends on radio conditions, timeslot allocation and network load.

3. Tools and Experimental Setup

Use a GSM/GPRS network simulator or a suitable teaching simulation. Model a mobile station, BTS/BSC, SGSN, GGSN and packet-data server. Configure available timeslots, coding scheme, signal quality, traffic load and background users.

4. Procedure

Create a packet-data session between the mobile station and server through the GPRS core.

Run a small data transfer and record throughput, packet delay and packet loss/BER if available.

Increase the number of active users and vary the timeslots allocated to the test user.

Change signal quality or coding scheme and repeat the transfer.

Introduce movement between cells and observe session continuity and mobility handling.

Compare lightly loaded and heavily loaded network results.

5. Observation

Parameter / Feature	**Expected observation**
Packet switching	Radio capacity is used when packets are sent, allowing statistical sharing among users.
Timeslots	More available timeslots can increase the user's data capacity, subject to network configuration.
Radio quality	Poor signal quality can require more robust coding and reduce effective data rate.
Network load	More simultaneous users can increase waiting

	time and reduce per-user throughput.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

GPRS is designed for packet data rather than a permanently dedicated circuit. Its throughput is modest compared with later 3G/4G/5G systems and varies with timeslot allocation, coding, signal quality and congestion. Mobility procedures help maintain packet service as a user changes cells, although interruptions may occur.

7. Optimization Techniques

Allocate timeslots according to demand, improve radio coverage, manage congestion, select appropriate coding schemes and maintain efficient mobility/session management.

8. Result / Conclusion

The experiment demonstrates packet-switched data delivery over GSM/GPRS. Performance depends on timeslot availability, coding scheme, radio quality and the number of active users.

QUESTION 16: Experiment based on HSCSD

1. Aim

To study High-Speed Circuit-Switched Data (HSCSD) and evaluate how combining GSM timeslots affects data performance.

2. Theory / Concept

HSCSD is a GSM enhancement that increases circuit-switched data capacity by assigning multiple timeslots to one connection.

Unlike packet-switched GPRS, HSCSD reserves circuit resources for the duration of the connection.

The achievable data rate depends on the number of allocated timeslots and the channel coding used.

3. Tools and Experimental Setup

Use a GSM/HSCSD teaching simulator. Configure a mobile station and base station, a circuit-switched data connection, available timeslots, coding rate, signal quality and competing users.

4. Procedure

Establish a baseline data connection using one timeslot and record throughput and delay.

Increase the allocated timeslots step by step, within the simulator's supported limit.

Measure the resulting data rate and resource usage for each allocation.

Add competing users and observe whether sufficient circuit resources remain available.

Reduce signal quality or add interference and repeat the measurements.

Compare HSCSD's reserved-resource behavior with packet-sharing behavior conceptually.

5. Observation

Parameter / Feature	**Expected observation**
Timeslot allocation	Allocating more timeslots increases the potential data rate for the connection.
Resource reservation	Reserved circuit resources may remain unavailable to other users even during idle periods.
Interference	Poor radio conditions can reduce effective performance or increase errors.
User capacity	A connection using several timeslots consumes more network resources.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

HSCSD can provide a higher circuit-switched data rate than a basic GSM data connection by combining timeslots. Its limitation is inefficient resource use for bursty data because resources are reserved for the session. The number of available timeslots and radio quality constrain performance.

7. Optimization Techniques

Choose the number of timeslots according to application demand, maintain good radio coverage and manage circuit-resource allocation to avoid excessive blocking.

8. Result / Conclusion

The experiment verifies that HSCSD increases circuit-switched data capacity through multi-timeslot allocation, but it uses dedicated resources and is less flexible for intermittent packet traffic than packet-switched services.

QUESTION 17: Experiment based on 1G

1. Aim

To examine first-generation (1G) mobile communication and evaluate its basic analog voice characteristics.

2. Theory / Concept

1G systems were the first generation of cellular mobile networks and used analog radio transmission mainly for voice.

Common 1G systems included AMPS, NMT and TACS in different regions.

Frequency Division Multiple Access (FDMA) was commonly used to assign separate frequency channels to calls.

1G had limited capacity, weak security and lower voice quality compared with digital generations.

3. Tools and Experimental Setup

Use a cellular-system simulator or a comparative model of an analog FDMA network. Configure a cell, frequency channels, transmit power, noise, path loss and multiple mobile users.

4. Procedure

Create a cell and assign a separate frequency channel to each active call.

Start one voice connection and record signal strength and voice-link quality.

Add users until the available channels are occupied; record blocking or call rejection.

Increase noise and co-channel interference and observe voice quality.

Move a mobile user toward the cell edge and observe signal degradation and handover behavior if modeled.

Compare the system's capacity, security and service type with later digital generations.

5. Observation

Parameter / Feature	**Expected observation**
Access method	FDMA assigns separate frequency channels to calls.
Service	Primarily analog voice; data services were very limited.
Capacity	Limited by the number of available radio channels.
Security and quality	Analog transmission offered weaker privacy and was more susceptible to noise.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As users increase, the finite number of frequency channels limits capacity. Noise and interference directly affect analog voice quality. 1G demonstrates the basic cellular concept but lacks the spectral efficiency, digital security and data capabilities of later generations.

7. Optimization Techniques

Within a legacy 1G model, careful frequency planning, suitable channel reuse and adequate coverage can reduce interference. The major long-term improvement is migration to digital cellular technologies.

8. Result / Conclusion

The experiment shows that 1G enabled mobile analog voice communication using cellular frequency channels, but its capacity, security and service capabilities were limited.

QUESTION 18: Experiment based on 2G

1. Aim

To study second-generation (2G) digital cellular communication and compare its performance with analog 1G systems.

2. Theory / Concept

2G introduced digital voice transmission and improved capacity, security and service quality.

GSM commonly uses TDMA with FDMA carrier allocation; IS-95 uses CDMA.

Digital systems support services such as SMS and, with enhancements, packet data.

Speech coding, channel coding and encryption improve efficiency and communication security.

3. Tools and Experimental Setup

Use a GSM/2G or equivalent cellular simulator. Configure a base station, mobile users, carrier channels, time slots, signal strength, noise and traffic. Select voice or SMS/data traffic.

4. Procedure

Configure a digital 2G cell and register a mobile user.

Establish a voice call or send a test message; verify successful delivery.

Add users and observe channel/time-slot allocation and call blocking.

Vary signal strength and interference; record BER, call quality and throughput for data traffic.

Introduce mobility and observe handover between cells.

Compare digital 2G performance with the analog 1G baseline.

5. Observation

Parameter / Feature	**Expected observation**
Digital transmission	Voice is represented digitally, allowing coding and error protection.
Capacity	TDMA/FDMA or CDMA methods support more organized sharing than basic analog systems.
Services	Voice and SMS are supported; packet-data enhancements include GPRS and EDGE.
Security	Digital authentication and encryption provide better protection than 1G.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Digital coding can improve voice quality when radio errors are within the system's correction capability. Under weak signal conditions, BER rises and service quality can fall. Capacity depends on the access method, channel allocation, interference and network configuration.

7. Optimization Techniques

Use proper frequency planning, power control, channel coding, interference management and efficient handover configuration.

8. Result / Conclusion

The experiment demonstrates that 2G improved mobile communication through digital transmission, better security, increased capacity and services such as SMS, with later enhancements supporting packet data.

QUESTION 19: Experiment based on 3G

1. Aim

To study third-generation (3G) mobile networks and evaluate their support for voice and mobile data.

2. Theory / Concept

3G networks were designed to provide higher data rates and multimedia services in addition to voice.

UMTS commonly uses WCDMA radio access; CDMA2000 is another 3G family.

3G supports services such as mobile internet, video calling and multimedia messaging.

Packet data and improved radio techniques increase capability compared with 2G.

3. Tools and Experimental Setup

Use a UMTS/WCDMA or 3G network simulator. Model a user equipment device, Node B, radio network controller/core network and a data server. Configure channel bandwidth, transmit power, noise, interference, mobility and traffic.

4. Procedure

Register a mobile device with the 3G network and establish a data session.

Run a web/data transfer and record throughput, delay and packet loss or BER.

Add multiple users and vary their traffic load.

Change signal strength, interference and mobility; repeat the measurements.

Test a voice or multimedia session if the simulator supports it.

Compare 3G results with 2G, keeping traffic and test conditions as similar as possible.

5. Observation

Parameter / Feature	**Expected observation**
Radio access	WCDMA spreads users over a wider carrier and separates them using codes.
Data service	Supports mobile internet and multimedia applications at higher rates than typical 2G services.
Interference	Cell loading and interference affect the shared WCDMA capacity.
Mobility	Handover and radio-resource control support users moving between cells.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

3G generally provides higher data capacity than 2G, but actual throughput depends on radio conditions, cell load, user mobility and service configuration. Increased interference and user count can reduce data rates and increase delay.

7. Optimization Techniques

Use power control, admission control, suitable spreading codes, load balancing, robust coding and well-planned handover parameters.

8. Result / Conclusion

The experiment shows that 3G extends mobile networks beyond basic voice and messaging by enabling practical mobile internet and multimedia services, with performance dependent on radio quality and cell loading.

QUESTION 20: Experiment based on 4G

1. Aim

To study fourth-generation (4G LTE) mobile communication and measure broadband data performance.

2. Theory / Concept

4G LTE is an all-IP mobile broadband system designed for high data rates, lower latency and efficient packet communication.

LTE uses OFDMA in the downlink and SC-FDMA in the uplink.

MIMO, adaptive modulation and coding, and flexible bandwidths improve spectral efficiency and capacity.

Voice can be provided using Voice over LTE (VoLTE) where deployed.

3. Tools and Experimental Setup

Use an LTE network simulator. Configure user equipment, eNodeB, EPC/core network and a server. Set channel bandwidth, resource blocks, modulation/coding, transmit power, noise, interference and mobility.

4. Procedure

Attach the user equipment to the LTE network and establish an IP data connection.

Run a data transfer and record throughput, average delay, packet loss and BER if available.

Increase the number of users and offered traffic; observe scheduling and resource sharing.

Change bandwidth, signal quality and interference one at a time.

Enable mobility and observe handover and data-session continuity.

Compare LTE results with 3G under equivalent traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
Downlink/uplink	LTE uses OFDMA downlink and SC-FDMA uplink.
Throughput	Broadband data rates are higher than typical 3G performance under suitable conditions.
Latency	An all-IP architecture and optimized radio procedures support lower latency.
Resource allocation	The scheduler assigns resource blocks according to channel quality and traffic demand.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

LTE performance improves with available bandwidth, good SINR, efficient scheduling and MIMO support. More users and interference compete for finite resources, which can lower per-user throughput and increase delay. Mobility requires handover between eNodeBs.

7. Optimization Techniques

Use channel-aware scheduling, MIMO, adaptive modulation/coding, interference coordination, load balancing and appropriate handover settings.

8. Result / Conclusion

The experiment demonstrates that 4G LTE provides high-speed IP-based mobile broadband with efficient OFDMA-based resource allocation and lower latency than earlier generations.

QUESTION 21: Experiment based on 5G

1. Aim

To study fifth-generation (5G) mobile communication and evaluate its high-speed, low-latency and massive-connectivity capabilities.

2. Theory / Concept

5G is designed to support enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC).

5G NR uses flexible OFDM-based radio technology, scalable numerology and advanced scheduling.

Massive MIMO and beamforming improve capacity and spatial coverage.

5G can operate in sub-6 GHz and millimeter-wave frequency ranges; higher frequencies can provide wide bandwidth but are more sensitive to blockage and propagation loss.

3. Tools and Experimental Setup

Use a 5G NR simulator or network simulation environment. Configure a gNodeB, user equipment, 5G core, carrier frequency, channel bandwidth, numerology, antenna configuration, traffic type, noise, interference and mobility.

4. Procedure

Configure a 5G cell and connect one user equipment device.

Run an eMBB data flow and record throughput, delay and packet loss/BER if supported.

Add users or IoT devices and observe resource allocation and per-user performance.

Test a low-latency traffic profile and measure end-to-end delay.

Vary frequency, bandwidth, beamforming, blockage and mobility; record the effect.

Compare the results with a 4G LTE baseline using comparable traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
eMBB	Supports high data rates for applications such as high-resolution streaming.
URLLC	Targets very low latency and high reliability for demanding control applications.
mMTC	Supports large numbers of connected devices with small, intermittent data transmissions.
Beamforming / MIMO	Can improve received signal quality and spatial reuse, depending on channel conditions.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

5G can achieve high throughput when wide bandwidth, good signal quality and suitable radio resources are available. Millimeter-wave links can be affected by blockage and short coverage range. Low latency depends on radio configuration, network architecture, load and application path; it is not guaranteed in every deployment.

7. Optimization Techniques

Use beam management, massive MIMO, suitable spectrum, edge computing for latency-sensitive services, QoS scheduling, interference coordination and dense small-cell deployment where justified.

8. Result / Conclusion

The experiment illustrates 5G's flexible support for broadband, low-latency and massive-IoT use cases. Actual performance depends on spectrum, deployment, signal conditions, network load and device capability.

QUESTION 22: Experiment based on Spread Spectrum (DSSS/FHSS)

1. Aim

To compare Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) under noise and interference.

2. Theory / Concept

Spread spectrum transmits a signal over a bandwidth wider than the minimum required for the information.

DSSS multiplies the data by a high-rate spreading code; the receiver uses the same code to despread the signal.

FHSS changes the carrier frequency according to a predefined hopping sequence known to transmitter and receiver.

Both methods can provide resistance to certain forms of interference, but their performance depends on codes, hopping pattern, synchronization and channel conditions.

3. Tools and Experimental Setup

Use a digital communication simulator. Configure a bit source, DSSS and FHSS transmitters/receivers, spreading code or hopping sequence, channel bandwidth, transmit power, AWGN noise and an interference source. Use the same data and comparable power for both tests.

4. Procedure

Generate a known binary data sequence and transmit it using DSSS.

At the receiver, synchronize the spreading code and despread the received signal.

Repeat using FHSS with a shared hopping sequence and synchronized transmitter/receiver.

Measure BER, throughput and occupied bandwidth with no added interference.

Add narrowband interference for one run and broadband noise for another; repeat the measurements.

Change the spreading factor or hopping rate and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
DSSS	Spreads each data symbol using a code; processing gain can reduce the effect of narrowband interference.
FHSS	Moves the carrier among frequencies; narrowband interference may affect only some hops.

Synchronization	Code or hopping-sequence mismatch prevents reliable recovery.
Bandwidth	Both methods use more spectrum than the unspread information signal.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

A higher spreading factor can improve processing gain but uses more chips per information bit and may reduce the net data rate for a fixed chip rate. FHSS performance depends on hop timing, number of channels and how often a hop encounters interference. Noise generally increases BER in both systems.

7. Optimization Techniques

Choose suitable spreading codes or hopping patterns, maintain synchronization, use appropriate transmit power and channel planning, and select spreading/hopping parameters according to interference conditions.

8. Result / Conclusion

The experiment shows that DSSS uses code-based spreading while FHSS uses frequency changes. Both can improve robustness against selected interference types, but neither eliminates all noise or interference.

QUESTION 23: Experiment based on Multiple Access (FDMA/TDMA/CDMA/OFDMA)

1. Aim

To compare FDMA, TDMA, CDMA and OFDMA when multiple users share a wireless channel.

2. Theory / Concept

FDMA assigns a separate frequency band to each user; guard bands reduce adjacent-channel interference.

TDMA assigns users different time slots on a shared carrier; synchronization is essential.

CDMA allows users to transmit over the same band using different spreading codes; code orthogonality and power control are important.

OFDMA divides the channel into orthogonal subcarriers and allocates groups of subcarriers to users. It is flexible for varying traffic.

3. Tools and Experimental Setup

Use a wireless-system simulator (for example, MATLAB/Simulink, ns-3 or an equivalent lab simulator). Configure four users, a common channel bandwidth, transmit power, receiver noise, path loss and an interference level. Keep the traffic load and channel conditions identical for a fair comparison.

4. Procedure

Create four user nodes and configure the same offered traffic for each user.

Run the simulation separately using FDMA, TDMA, CDMA and OFDMA.

Record bandwidth allocation, received signal strength, throughput, average delay, bit error rate (BER) and spectral efficiency.

Increase the number of users and offered load in stages; repeat the measurements.

Increase noise/interference and then introduce mobility or changing path loss. Record the effect on each technique.

Plot throughput, delay and BER against user count or signal-to-noise ratio (SNR), and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
FDMA	Fixed frequency allocation; guard bands consume some bandwidth; performance can be predictable at low load.
TDMA	Time-slot sharing; delay can rise when users wait for their assigned slots; timing synchronization is

	required.
CDMA	Users share time and frequency; interference rises as users increase; power control helps limit the near-far problem.
OFDMA	Subcarriers can be allocated dynamically; generally supports efficient scheduling and high data rates in broadband systems.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As the number of active users and interference increase, shared resources become more heavily loaded. FDMA may waste capacity when a user is idle; TDMA can add waiting time; CDMA can become interference-limited; OFDMA can adapt resource blocks to traffic but requires accurate synchronization and channel estimation. Higher noise generally increases BER and retransmissions, reducing useful throughput.

7. Optimization Techniques

Use adaptive modulation and coding, efficient scheduling, power control, channel estimation, interference coordination and suitable guard intervals. In a mobile environment, use handover and channel tracking where supported.

8. Result / Conclusion

The experiment demonstrates that no single multiple-access method is best for every condition. OFDMA offers flexible resource allocation for broadband traffic, while FDMA, TDMA and CDMA illustrate frequency-, time- and code-based sharing. The best choice depends on user density, mobility, interference, delay and bandwidth requirements.

QUESTION 24: Experiment based on Radio Interfaces

1. Aim

To study a cellular radio interface and evaluate how its radio parameters affect communication performance.

2. Theory / Concept

A radio interface defines how a mobile device communicates over the air with a base station, including the physical channel, modulation, coding, access method and radio-resource procedures.

Important configurable parameters include carrier frequency, channel bandwidth, transmit power, modulation, coding rate, noise and interference.

The received signal quality is affected by path loss, fading, shadowing, obstacles and user mobility.

3. Tools and Experimental Setup

Use a cellular radio-interface simulator or a link-level/network simulator. Configure a mobile station, base station, carrier frequency, channel bandwidth, transmit power, receiver noise and a traffic source. Select a supported modulation and coding scheme.

4. Procedure

Establish a baseline link with one stationary user and record received signal strength and SNR.

Generate data traffic and measure throughput, delay and BER.

Change bandwidth while keeping other settings constant; record the change in throughput and spectral efficiency.

Vary transmit power and receiver noise separately, noting the effect on SNR and BER.

Add a second and then additional users to observe resource sharing and interference.

Introduce mobility and obstacles/fading in the simulator; observe signal variation and any handover or link adaptation.

5. Observation

Parameter / Feature	**Expected observation**
Bandwidth	A wider channel can support a higher peak data rate, subject to radio conditions and system limits.
Signal strength / SNR	A stronger desired signal relative to noise generally improves decoding and lowers BER.
Interference	Co-channel or adjacent-channel interference can reduce SINR, throughput and reliability.
Mobility	Movement changes path loss and fading; link

	adaptation or handover may be needed.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput is the successfully delivered data per unit time. Average delay is the time taken for packets to reach the destination. BER is erroneous bits divided by total transmitted bits. Spectral efficiency is throughput per unit bandwidth, commonly expressed in bit/s/Hz. The results should show that radio quality, bandwidth, coding and user load jointly determine performance.

7. Optimization Techniques

Apply power control, adaptive modulation/coding, robust channel coding, interference coordination, antenna diversity/beamforming where available, and appropriate handover thresholds.

8. Result / Conclusion

The radio-interface experiment shows how physical-layer settings and radio conditions affect throughput, delay, BER and spectral efficiency. Reliable performance requires balancing data rate, coverage, interference and mobility.

QUESTION 25: Experiment based on Protocols

1. Aim

To analyze the role of mobile-network protocols in reliable data communication and measure their performance under changing wireless conditions.

2. Theory / Concept

Protocols define the rules for addressing, packet formatting, transmission, error handling, routing and delivery.

A mobile data path commonly involves application, transport, network and link/physical functions.

TCP provides reliable, ordered delivery using acknowledgements and retransmissions; UDP has lower protocol overhead but does not itself guarantee delivery.

Wireless losses, variable delay and mobility can affect protocol performance.

3. Tools and Experimental Setup

Use a network simulator such as ns-3 or an equivalent tool. Create a mobile node, an access point/base station, a server and a traffic application. Configure a wireless link, IP addressing and either TCP or UDP traffic.

4. Procedure

Build the topology and verify that the mobile node can reach the server.

Run a baseline TCP flow and record throughput, packet delivery, delay and packet loss.

Repeat using UDP with the same offered traffic rate.

Increase background traffic and the number of mobile users; record congestion effects.

Add wireless errors, higher delay and mobility; observe retransmissions, packet loss and throughput.

Compare TCP and UDP results and explain the effect of reliability mechanisms.

5. Observation

Parameter / Feature	**Expected observation**
TCP	Acknowledgements and retransmissions improve reliable delivery, but can increase delay and reduce throughput on lossy links.
UDP	Lower overhead and no built-in retransmission; delay may be lower, but the application must tolerate or handle loss.
Congestion	Higher offered load can cause queueing delay, packet drops and reduced goodput.

Mobility	Route changes or temporary disconnections can interrupt flows and trigger recovery.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput/goodput measures useful delivered data. Delay includes transmission, propagation, processing and queueing time. Packet loss and retransmissions become more significant under congestion or weak radio conditions. TCP is suitable where complete ordered delivery is important; UDP is useful for delay-sensitive real-time traffic when the application can manage losses.

7. Optimization Techniques

Use suitable transport protocols, congestion control, traffic prioritization, buffering, robust wireless links, mobility-aware routing and application-level error handling where needed.

8. Result / Conclusion

The experiment highlights that protocol selection affects reliability, delay and throughput. TCP prioritizes reliable ordered delivery, while UDP offers lower overhead and is often selected for real-time applications with suitable loss handling.

QUESTION 26: Experiment based on GPRS

1. Aim

To study General Packet Radio Service (GPRS), a packet-switched data service introduced for GSM networks, and evaluate its performance.

2. Theory / Concept

GPRS adds packet-data capability to GSM and allows radio resources to be shared by users when data is sent.

Its network architecture includes the Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN), which support mobility/session management and connection to external packet networks.

GPRS uses allocated GSM timeslots and coding schemes; actual data rate depends on radio conditions, timeslot allocation and network load.

3. Tools and Experimental Setup

Use a GSM/GPRS network simulator or a suitable teaching simulation. Model a mobile station, BTS/BSC, SGSN, GGSN and packet-data server. Configure available timeslots, coding scheme, signal quality, traffic load and background users.

4. Procedure

Create a packet-data session between the mobile station and server through the GPRS core.

Run a small data transfer and record throughput, packet delay and packet loss/BER if available.

Increase the number of active users and vary the timeslots allocated to the test user.

Change signal quality or coding scheme and repeat the transfer.

Introduce movement between cells and observe session continuity and mobility handling.

Compare lightly loaded and heavily loaded network results.

5. Observation

Parameter / Feature	**Expected observation**
Packet switching	Radio capacity is used when packets are sent, allowing statistical sharing among users.
Timeslots	More available timeslots can increase the user's data capacity, subject to network configuration.
Radio quality	Poor signal quality can require more robust coding and reduce effective data rate.
Network load	More simultaneous users can increase waiting

	time and reduce per-user throughput.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

GPRS is designed for packet data rather than a permanently dedicated circuit. Its throughput is modest compared with later 3G/4G/5G systems and varies with timeslot allocation, coding, signal quality and congestion. Mobility procedures help maintain packet service as a user changes cells, although interruptions may occur.

7. Optimization Techniques

Allocate timeslots according to demand, improve radio coverage, manage congestion, select appropriate coding schemes and maintain efficient mobility/session management.

8. Result / Conclusion

The experiment demonstrates packet-switched data delivery over GSM/GPRS. Performance depends on timeslot availability, coding scheme, radio quality and the number of active users.

QUESTION 27: Experiment based on HSCSD

1. Aim

To study High-Speed Circuit-Switched Data (HSCSD) and evaluate how combining GSM timeslots affects data performance.

2. Theory / Concept

HSCSD is a GSM enhancement that increases circuit-switched data capacity by assigning multiple timeslots to one connection.

Unlike packet-switched GPRS, HSCSD reserves circuit resources for the duration of the connection.

The achievable data rate depends on the number of allocated timeslots and the channel coding used.

3. Tools and Experimental Setup

Use a GSM/HSCSD teaching simulator. Configure a mobile station and base station, a circuit-switched data connection, available timeslots, coding rate, signal quality and competing users.

4. Procedure

Establish a baseline data connection using one timeslot and record throughput and delay.

Increase the allocated timeslots step by step, within the simulator's supported limit.

Measure the resulting data rate and resource usage for each allocation.

Add competing users and observe whether sufficient circuit resources remain available.

Reduce signal quality or add interference and repeat the measurements.

Compare HSCSD's reserved-resource behavior with packet-sharing behavior conceptually.

5. Observation

Parameter / Feature	**Expected observation**
Timeslot allocation	Allocating more timeslots increases the potential data rate for the connection.
Resource reservation	Reserved circuit resources may remain unavailable to other users even during idle periods.
Interference	Poor radio conditions can reduce effective performance or increase errors.
User capacity	A connection using several timeslots consumes more network resources.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

HSCSD can provide a higher circuit-switched data rate than a basic GSM data connection by combining timeslots. Its limitation is inefficient resource use for bursty data because resources are reserved for the session. The number of available timeslots and radio quality constrain performance.

7. Optimization Techniques

Choose the number of timeslots according to application demand, maintain good radio coverage and manage circuit-resource allocation to avoid excessive blocking.

8. Result / Conclusion

The experiment verifies that HSCSD increases circuit-switched data capacity through multi-timeslot allocation, but it uses dedicated resources and is less flexible for intermittent packet traffic than packet-switched services.

QUESTION 28: Experiment based on 1G

1. Aim

To examine first-generation (1G) mobile communication and evaluate its basic analog voice characteristics.

2. Theory / Concept

1G systems were the first generation of cellular mobile networks and used analog radio transmission mainly for voice.

Common 1G systems included AMPS, NMT and TACS in different regions.

Frequency Division Multiple Access (FDMA) was commonly used to assign separate frequency channels to calls.

1G had limited capacity, weak security and lower voice quality compared with digital generations.

3. Tools and Experimental Setup

Use a cellular-system simulator or a comparative model of an analog FDMA network. Configure a cell, frequency channels, transmit power, noise, path loss and multiple mobile users.

4. Procedure

Create a cell and assign a separate frequency channel to each active call.

Start one voice connection and record signal strength and voice-link quality.

Add users until the available channels are occupied; record blocking or call rejection.

Increase noise and co-channel interference and observe voice quality.

Move a mobile user toward the cell edge and observe signal degradation and handover behavior if modeled.

Compare the system's capacity, security and service type with later digital generations.

5. Observation

Parameter / Feature	**Expected observation**
Access method	FDMA assigns separate frequency channels to calls.
Service	Primarily analog voice; data services were very limited.
Capacity	Limited by the number of available radio channels.
Security and quality	Analog transmission offered weaker privacy and was more susceptible to noise.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As users increase, the finite number of frequency channels limits capacity. Noise and interference directly affect analog voice quality. 1G demonstrates the basic cellular concept but lacks the spectral efficiency, digital security and data capabilities of later generations.

7. Optimization Techniques

Within a legacy 1G model, careful frequency planning, suitable channel reuse and adequate coverage can reduce interference. The major long-term improvement is migration to digital cellular technologies.

8. Result / Conclusion

The experiment shows that 1G enabled mobile analog voice communication using cellular frequency channels, but its capacity, security and service capabilities were limited.

QUESTION 29: Experiment based on 2G

1. Aim

To study second-generation (2G) digital cellular communication and compare its performance with analog 1G systems.

2. Theory / Concept

2G introduced digital voice transmission and improved capacity, security and service quality.

GSM commonly uses TDMA with FDMA carrier allocation; IS-95 uses CDMA.

Digital systems support services such as SMS and, with enhancements, packet data.

Speech coding, channel coding and encryption improve efficiency and communication security.

3. Tools and Experimental Setup

Use a GSM/2G or equivalent cellular simulator. Configure a base station, mobile users, carrier channels, time slots, signal strength, noise and traffic. Select voice or SMS/data traffic.

4. Procedure

Configure a digital 2G cell and register a mobile user.

Establish a voice call or send a test message; verify successful delivery.

Add users and observe channel/time-slot allocation and call blocking.

Vary signal strength and interference; record BER, call quality and throughput for data traffic.

Introduce mobility and observe handover between cells.

Compare digital 2G performance with the analog 1G baseline.

5. Observation

Parameter / Feature	**Expected observation**
Digital transmission	Voice is represented digitally, allowing coding and error protection.
Capacity	TDMA/FDMA or CDMA methods support more organized sharing than basic analog systems.
Services	Voice and SMS are supported; packet-data enhancements include GPRS and EDGE.
Security	Digital authentication and encryption provide better protection than 1G.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Digital coding can improve voice quality when radio errors are within the system's correction capability. Under weak signal conditions, BER rises and service quality can fall. Capacity depends on the access method, channel allocation, interference and network configuration.

7. Optimization Techniques

Use proper frequency planning, power control, channel coding, interference management and efficient handover configuration.

8. Result / Conclusion

The experiment demonstrates that 2G improved mobile communication through digital transmission, better security, increased capacity and services such as SMS, with later enhancements supporting packet data.

QUESTION 30: Experiment based on 3G

1. Aim

To study third-generation (3G) mobile networks and evaluate their support for voice and mobile data.

2. Theory / Concept

3G networks were designed to provide higher data rates and multimedia services in addition to voice.

UMTS commonly uses WCDMA radio access; CDMA2000 is another 3G family.

3G supports services such as mobile internet, video calling and multimedia messaging.

Packet data and improved radio techniques increase capability compared with 2G.

3. Tools and Experimental Setup

Use a UMTS/WCDMA or 3G network simulator. Model a user equipment device, Node B, radio network controller/core network and a data server. Configure channel bandwidth, transmit power, noise, interference, mobility and traffic.

4. Procedure

Register a mobile device with the 3G network and establish a data session.

Run a web/data transfer and record throughput, delay and packet loss or BER.

Add multiple users and vary their traffic load.

Change signal strength, interference and mobility; repeat the measurements.

Test a voice or multimedia session if the simulator supports it.

Compare 3G results with 2G, keeping traffic and test conditions as similar as possible.

5. Observation

Parameter / Feature	**Expected observation**
Radio access	WCDMA spreads users over a wider carrier and separates them using codes.
Data service	Supports mobile internet and multimedia applications at higher rates than typical 2G services.
Interference	Cell loading and interference affect the shared WCDMA capacity.
Mobility	Handover and radio-resource control support users moving between cells.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

3G generally provides higher data capacity than 2G, but actual throughput depends on radio conditions, cell load, user mobility and service configuration. Increased interference and user count can reduce data rates and increase delay.

7. Optimization Techniques

Use power control, admission control, suitable spreading codes, load balancing, robust coding and well-planned handover parameters.

8. Result / Conclusion

The experiment shows that 3G extends mobile networks beyond basic voice and messaging by enabling practical mobile internet and multimedia services, with performance dependent on radio quality and cell loading.

QUESTION 31: Experiment based on 4G

1. Aim

To study fourth-generation (4G LTE) mobile communication and measure broadband data performance.

2. Theory / Concept

4G LTE is an all-IP mobile broadband system designed for high data rates, lower latency and efficient packet communication.

LTE uses OFDMA in the downlink and SC-FDMA in the uplink.

MIMO, adaptive modulation and coding, and flexible bandwidths improve spectral efficiency and capacity.

Voice can be provided using Voice over LTE (VoLTE) where deployed.

3. Tools and Experimental Setup

Use an LTE network simulator. Configure user equipment, eNodeB, EPC/core network and a server. Set channel bandwidth, resource blocks, modulation/coding, transmit power, noise, interference and mobility.

4. Procedure

Attach the user equipment to the LTE network and establish an IP data connection.

Run a data transfer and record throughput, average delay, packet loss and BER if available.

Increase the number of users and offered traffic; observe scheduling and resource sharing.

Change bandwidth, signal quality and interference one at a time.

Enable mobility and observe handover and data-session continuity.

Compare LTE results with 3G under equivalent traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
Downlink/uplink	LTE uses OFDMA downlink and SC-FDMA uplink.
Throughput	Broadband data rates are higher than typical 3G performance under suitable conditions.
Latency	An all-IP architecture and optimized radio procedures support lower latency.
Resource allocation	The scheduler assigns resource blocks according to channel quality and traffic demand.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

LTE performance improves with available bandwidth, good SINR, efficient scheduling and MIMO support. More users and interference compete for finite resources, which can lower per-user throughput and increase delay. Mobility requires handover between eNodeBs.

7. Optimization Techniques

Use channel-aware scheduling, MIMO, adaptive modulation/coding, interference coordination, load balancing and appropriate handover settings.

8. Result / Conclusion

The experiment demonstrates that 4G LTE provides high-speed IP-based mobile broadband with efficient OFDMA-based resource allocation and lower latency than earlier generations.

QUESTION 32: Experiment based on 5G

1. Aim

To study fifth-generation (5G) mobile communication and evaluate its high-speed, low-latency and massive-connectivity capabilities.

2. Theory / Concept

5G is designed to support enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC).

5G NR uses flexible OFDM-based radio technology, scalable numerology and advanced scheduling.

Massive MIMO and beamforming improve capacity and spatial coverage.

5G can operate in sub-6 GHz and millimeter-wave frequency ranges; higher frequencies can provide wide bandwidth but are more sensitive to blockage and propagation loss.

3. Tools and Experimental Setup

Use a 5G NR simulator or network simulation environment. Configure a gNodeB, user equipment, 5G core, carrier frequency, channel bandwidth, numerology, antenna configuration, traffic type, noise, interference and mobility.

4. Procedure

Configure a 5G cell and connect one user equipment device.

Run an eMBB data flow and record throughput, delay and packet loss/BER if supported.

Add users or IoT devices and observe resource allocation and per-user performance.

Test a low-latency traffic profile and measure end-to-end delay.

Vary frequency, bandwidth, beamforming, blockage and mobility; record the effect.

Compare the results with a 4G LTE baseline using comparable traffic and channel conditions.

5. Observation

Parameter / Feature	**Expected observation**
eMBB	Supports high data rates for applications such as high-resolution streaming.
URLLC	Targets very low latency and high reliability for demanding control applications.
mMTC	Supports large numbers of connected devices with small, intermittent data transmissions.
Beamforming / MIMO	Can improve received signal quality and spatial reuse, depending on channel conditions.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

5G can achieve high throughput when wide bandwidth, good signal quality and suitable radio resources are available. Millimeter-wave links can be affected by blockage and short coverage range. Low latency depends on radio configuration, network architecture, load and application path; it is not guaranteed in every deployment.

7. Optimization Techniques

Use beam management, massive MIMO, suitable spectrum, edge computing for latency-sensitive services, QoS scheduling, interference coordination and dense small-cell deployment where justified.

8. Result / Conclusion

The experiment illustrates 5G's flexible support for broadband, low-latency and massive-IoT use cases. Actual performance depends on spectrum, deployment, signal conditions, network load and device capability.

QUESTION 33: Experiment based on Spread Spectrum (DSSS/FHSS)

1. Aim

To compare Direct Sequence Spread Spectrum (DSSS) and Frequency Hopping Spread Spectrum (FHSS) under noise and interference.

2. Theory / Concept

Spread spectrum transmits a signal over a bandwidth wider than the minimum required for the information.

DSSS multiplies the data by a high-rate spreading code; the receiver uses the same code to despread the signal.

FHSS changes the carrier frequency according to a predefined hopping sequence known to transmitter and receiver.

Both methods can provide resistance to certain forms of interference, but their performance depends on codes, hopping pattern, synchronization and channel conditions.

3. Tools and Experimental Setup

Use a digital communication simulator. Configure a bit source, DSSS and FHSS transmitters/receivers, spreading code or hopping sequence, channel bandwidth, transmit power, AWGN noise and an interference source. Use the same data and comparable power for both tests.

4. Procedure

Generate a known binary data sequence and transmit it using DSSS.

At the receiver, synchronize the spreading code and despread the received signal.

Repeat using FHSS with a shared hopping sequence and synchronized transmitter/receiver.

Measure BER, throughput and occupied bandwidth with no added interference.

Add narrowband interference for one run and broadband noise for another; repeat the measurements.

Change the spreading factor or hopping rate and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
DSSS	Spreads each data symbol using a code; processing gain can reduce the effect of narrowband interference.
FHSS	Moves the carrier among frequencies; narrowband interference may affect only some hops.

Synchronization	Code or hopping-sequence mismatch prevents reliable recovery.
Bandwidth	Both methods use more spectrum than the unspread information signal.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

A higher spreading factor can improve processing gain but uses more chips per information bit and may reduce the net data rate for a fixed chip rate. FHSS performance depends on hop timing, number of channels and how often a hop encounters interference. Noise generally increases BER in both systems.

7. Optimization Techniques

Choose suitable spreading codes or hopping patterns, maintain synchronization, use appropriate transmit power and channel planning, and select spreading/hopping parameters according to interference conditions.

8. Result / Conclusion

The experiment shows that DSSS uses code-based spreading while FHSS uses frequency changes. Both can improve robustness against selected interference types, but neither eliminates all noise or interference.

QUESTION 34: Experiment based on Multiple Access (FDMA/TDMA/CDMA/OFDMA)

1. Aim

To compare FDMA, TDMA, CDMA and OFDMA when multiple users share a wireless channel.

2. Theory / Concept

FDMA assigns a separate frequency band to each user; guard bands reduce adjacent-channel interference.

TDMA assigns users different time slots on a shared carrier; synchronization is essential.

CDMA allows users to transmit over the same band using different spreading codes; code orthogonality and power control are important.

OFDMA divides the channel into orthogonal subcarriers and allocates groups of subcarriers to users. It is flexible for varying traffic.

3. Tools and Experimental Setup

Use a wireless-system simulator (for example, MATLAB/Simulink, ns-3 or an equivalent lab simulator). Configure four users, a common channel bandwidth, transmit power, receiver noise, path loss and an interference level. Keep the traffic load and channel conditions identical for a fair comparison.

4. Procedure

Create four user nodes and configure the same offered traffic for each user.

Run the simulation separately using FDMA, TDMA, CDMA and OFDMA.

Record bandwidth allocation, received signal strength, throughput, average delay, bit error rate (BER) and spectral efficiency.

Increase the number of users and offered load in stages; repeat the measurements.

Increase noise/interference and then introduce mobility or changing path loss. Record the effect on each technique.

Plot throughput, delay and BER against user count or signal-to-noise ratio (SNR), and compare the results.

5. Observation

Parameter / Feature	**Expected observation**
FDMA	Fixed frequency allocation; guard bands consume some bandwidth; performance can be predictable at low load.
TDMA	Time-slot sharing; delay can rise when users wait for their assigned slots; timing synchronization is

	required.
CDMA	Users share time and frequency; interference rises as users increase; power control helps limit the near-far problem.
OFDMA	Subcarriers can be allocated dynamically; generally supports efficient scheduling and high data rates in broadband systems.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As the number of active users and interference increase, shared resources become more heavily loaded. FDMA may waste capacity when a user is idle; TDMA can add waiting time; CDMA can become interference-limited; OFDMA can adapt resource blocks to traffic but requires accurate synchronization and channel estimation. Higher noise generally increases BER and retransmissions, reducing useful throughput.

7. Optimization Techniques

Use adaptive modulation and coding, efficient scheduling, power control, channel estimation, interference coordination and suitable guard intervals. In a mobile environment, use handover and channel tracking where supported.

8. Result / Conclusion

The experiment demonstrates that no single multiple-access method is best for every condition. OFDMA offers flexible resource allocation for broadband traffic, while FDMA, TDMA and CDMA illustrate frequency-, time- and code-based sharing. The best choice depends on user density, mobility, interference, delay and bandwidth requirements.

QUESTION 35: Experiment based on Radio Interfaces

1. Aim

To study a cellular radio interface and evaluate how its radio parameters affect communication performance.

2. Theory / Concept

A radio interface defines how a mobile device communicates over the air with a base station, including the physical channel, modulation, coding, access method and radio-resource procedures.

Important configurable parameters include carrier frequency, channel bandwidth, transmit power, modulation, coding rate, noise and interference.

The received signal quality is affected by path loss, fading, shadowing, obstacles and user mobility.

3. Tools and Experimental Setup

Use a cellular radio-interface simulator or a link-level/network simulator. Configure a mobile station, base station, carrier frequency, channel bandwidth, transmit power, receiver noise and a traffic source. Select a supported modulation and coding scheme.

4. Procedure

Establish a baseline link with one stationary user and record received signal strength and SNR.

Generate data traffic and measure throughput, delay and BER.

Change bandwidth while keeping other settings constant; record the change in throughput and spectral efficiency.

Vary transmit power and receiver noise separately, noting the effect on SNR and BER.

Add a second and then additional users to observe resource sharing and interference.

Introduce mobility and obstacles/fading in the simulator; observe signal variation and any handover or link adaptation.

5. Observation

Parameter / Feature	**Expected observation**
Bandwidth	A wider channel can support a higher peak data rate, subject to radio conditions and system limits.
Signal strength / SNR	A stronger desired signal relative to noise generally improves decoding and lowers BER.
Interference	Co-channel or adjacent-channel interference can reduce SINR, throughput and reliability.
Mobility	Movement changes path loss and fading; link

	adaptation or handover may be needed.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput is the successfully delivered data per unit time. Average delay is the time taken for packets to reach the destination. BER is erroneous bits divided by total transmitted bits. Spectral efficiency is throughput per unit bandwidth, commonly expressed in bit/s/Hz. The results should show that radio quality, bandwidth, coding and user load jointly determine performance.

7. Optimization Techniques

Apply power control, adaptive modulation/coding, robust channel coding, interference coordination, antenna diversity/beamforming where available, and appropriate handover thresholds.

8. Result / Conclusion

The radio-interface experiment shows how physical-layer settings and radio conditions affect throughput, delay, BER and spectral efficiency. Reliable performance requires balancing data rate, coverage, interference and mobility.

QUESTION 36: Experiment based on Protocols

1. Aim

To analyze the role of mobile-network protocols in reliable data communication and measure their performance under changing wireless conditions.

2. Theory / Concept

Protocols define the rules for addressing, packet formatting, transmission, error handling, routing and delivery.

A mobile data path commonly involves application, transport, network and link/physical functions.

TCP provides reliable, ordered delivery using acknowledgements and retransmissions; UDP has lower protocol overhead but does not itself guarantee delivery.

Wireless losses, variable delay and mobility can affect protocol performance.

3. Tools and Experimental Setup

Use a network simulator such as ns-3 or an equivalent tool. Create a mobile node, an access point/base station, a server and a traffic application. Configure a wireless link, IP addressing and either TCP or UDP traffic.

4. Procedure

Build the topology and verify that the mobile node can reach the server.

Run a baseline TCP flow and record throughput, packet delivery, delay and packet loss.

Repeat using UDP with the same offered traffic rate.

Increase background traffic and the number of mobile users; record congestion effects.

Add wireless errors, higher delay and mobility; observe retransmissions, packet loss and throughput.

Compare TCP and UDP results and explain the effect of reliability mechanisms.

5. Observation

Parameter / Feature	**Expected observation**
TCP	Acknowledgements and retransmissions improve reliable delivery, but can increase delay and reduce throughput on lossy links.
UDP	Lower overhead and no built-in retransmission; delay may be lower, but the application must tolerate or handle loss.
Congestion	Higher offered load can cause queueing delay, packet drops and reduced goodput.

Mobility	Route changes or temporary disconnections can interrupt flows and trigger recovery.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Throughput/goodput measures useful delivered data. Delay includes transmission, propagation, processing and queueing time. Packet loss and retransmissions become more significant under congestion or weak radio conditions. TCP is suitable where complete ordered delivery is important; UDP is useful for delay-sensitive real-time traffic when the application can manage losses.

7. Optimization Techniques

Use suitable transport protocols, congestion control, traffic prioritization, buffering, robust wireless links, mobility-aware routing and application-level error handling where needed.

8. Result / Conclusion

The experiment highlights that protocol selection affects reliability, delay and throughput. TCP prioritizes reliable ordered delivery, while UDP offers lower overhead and is often selected for real-time applications with suitable loss handling.

QUESTION 37: Experiment based on GPRS

1. Aim

To study General Packet Radio Service (GPRS), a packet-switched data service introduced for GSM networks, and evaluate its performance.

2. Theory / Concept

GPRS adds packet-data capability to GSM and allows radio resources to be shared by users when data is sent.

Its network architecture includes the Serving GPRS Support Node (SGSN) and Gateway GPRS Support Node (GGSN), which support mobility/session management and connection to external packet networks.

GPRS uses allocated GSM timeslots and coding schemes; actual data rate depends on radio conditions, timeslot allocation and network load.

3. Tools and Experimental Setup

Use a GSM/GPRS network simulator or a suitable teaching simulation. Model a mobile station, BTS/BSC, SGSN, GGSN and packet-data server. Configure available timeslots, coding scheme, signal quality, traffic load and background users.

4. Procedure

Create a packet-data session between the mobile station and server through the GPRS core.

Run a small data transfer and record throughput, packet delay and packet loss/BER if available.

Increase the number of active users and vary the timeslots allocated to the test user.

Change signal quality or coding scheme and repeat the transfer.

Introduce movement between cells and observe session continuity and mobility handling.

Compare lightly loaded and heavily loaded network results.

5. Observation

Parameter / Feature	**Expected observation**
Packet switching	Radio capacity is used when packets are sent, allowing statistical sharing among users.
Timeslots	More available timeslots can increase the user's data capacity, subject to network configuration.
Radio quality	Poor signal quality can require more robust coding and reduce effective data rate.
Network load	More simultaneous users can increase waiting

	time and reduce per-user throughput.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

GPRS is designed for packet data rather than a permanently dedicated circuit. Its throughput is modest compared with later 3G/4G/5G systems and varies with timeslot allocation, coding, signal quality and congestion. Mobility procedures help maintain packet service as a user changes cells, although interruptions may occur.

7. Optimization Techniques

Allocate timeslots according to demand, improve radio coverage, manage congestion, select appropriate coding schemes and maintain efficient mobility/session management.

8. Result / Conclusion

The experiment demonstrates packet-switched data delivery over GSM/GPRS. Performance depends on timeslot availability, coding scheme, radio quality and the number of active users.

QUESTION 38: Experiment based on HSCSD

1. Aim

To study High-Speed Circuit-Switched Data (HSCSD) and evaluate how combining GSM timeslots affects data performance.

2. Theory / Concept

HSCSD is a GSM enhancement that increases circuit-switched data capacity by assigning multiple timeslots to one connection.

Unlike packet-switched GPRS, HSCSD reserves circuit resources for the duration of the connection.

The achievable data rate depends on the number of allocated timeslots and the channel coding used.

3. Tools and Experimental Setup

Use a GSM/HSCSD teaching simulator. Configure a mobile station and base station, a circuit-switched data connection, available timeslots, coding rate, signal quality and competing users.

4. Procedure

Establish a baseline data connection using one timeslot and record throughput and delay.

Increase the allocated timeslots step by step, within the simulator's supported limit.

Measure the resulting data rate and resource usage for each allocation.

Add competing users and observe whether sufficient circuit resources remain available.

Reduce signal quality or add interference and repeat the measurements.

Compare HSCSD's reserved-resource behavior with packet-sharing behavior conceptually.

5. Observation

Parameter / Feature	**Expected observation**
Timeslot allocation	Allocating more timeslots increases the potential data rate for the connection.
Resource reservation	Reserved circuit resources may remain unavailable to other users even during idle periods.
Interference	Poor radio conditions can reduce effective performance or increase errors.
User capacity	A connection using several timeslots consumes more network resources.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

HSCSD can provide a higher circuit-switched data rate than a basic GSM data connection by combining timeslots. Its limitation is inefficient resource use for bursty data because resources are reserved for the session. The number of available timeslots and radio quality constrain performance.

7. Optimization Techniques

Choose the number of timeslots according to application demand, maintain good radio coverage and manage circuit-resource allocation to avoid excessive blocking.

8. Result / Conclusion

The experiment verifies that HSCSD increases circuit-switched data capacity through multi-timeslot allocation, but it uses dedicated resources and is less flexible for intermittent packet traffic than packet-switched services.

QUESTION 39: Experiment based on 1G

1. Aim

To examine first-generation (1G) mobile communication and evaluate its basic analog voice characteristics.

2. Theory / Concept

1G systems were the first generation of cellular mobile networks and used analog radio transmission mainly for voice.

Common 1G systems included AMPS, NMT and TACS in different regions.

Frequency Division Multiple Access (FDMA) was commonly used to assign separate frequency channels to calls.

1G had limited capacity, weak security and lower voice quality compared with digital generations.

3. Tools and Experimental Setup

Use a cellular-system simulator or a comparative model of an analog FDMA network. Configure a cell, frequency channels, transmit power, noise, path loss and multiple mobile users.

4. Procedure

Create a cell and assign a separate frequency channel to each active call.

Start one voice connection and record signal strength and voice-link quality.

Add users until the available channels are occupied; record blocking or call rejection.

Increase noise and co-channel interference and observe voice quality.

Move a mobile user toward the cell edge and observe signal degradation and handover behavior if modeled.

Compare the system's capacity, security and service type with later digital generations.

5. Observation

Parameter / Feature	**Expected observation**
Access method	FDMA assigns separate frequency channels to calls.
Service	Primarily analog voice; data services were very limited.
Capacity	Limited by the number of available radio channels.
Security and quality	Analog transmission offered weaker privacy and was more susceptible to noise.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

As users increase, the finite number of frequency channels limits capacity. Noise and interference directly affect analog voice quality. 1G demonstrates the basic cellular concept but lacks the spectral efficiency, digital security and data capabilities of later generations.

7. Optimization Techniques

Within a legacy 1G model, careful frequency planning, suitable channel reuse and adequate coverage can reduce interference. The major long-term improvement is migration to digital cellular technologies.

8. Result / Conclusion

The experiment shows that 1G enabled mobile analog voice communication using cellular frequency channels, but its capacity, security and service capabilities were limited.

QUESTION 40: Experiment based on 2G

1. Aim

To study second-generation (2G) digital cellular communication and compare its performance with analog 1G systems.

2. Theory / Concept

2G introduced digital voice transmission and improved capacity, security and service quality.

GSM commonly uses TDMA with FDMA carrier allocation; IS-95 uses CDMA.

Digital systems support services such as SMS and, with enhancements, packet data.

Speech coding, channel coding and encryption improve efficiency and communication security.

3. Tools and Experimental Setup

Use a GSM/2G or equivalent cellular simulator. Configure a base station, mobile users, carrier channels, time slots, signal strength, noise and traffic. Select voice or SMS/data traffic.

4. Procedure

Configure a digital 2G cell and register a mobile user.

Establish a voice call or send a test message; verify successful delivery.

Add users and observe channel/time-slot allocation and call blocking.

Vary signal strength and interference; record BER, call quality and throughput for data traffic.

Introduce mobility and observe handover between cells.

Compare digital 2G performance with the analog 1G baseline.

5. Observation

Parameter / Feature	**Expected observation**
Digital transmission	Voice is represented digitally, allowing coding and error protection.
Capacity	TDMA/FDMA or CDMA methods support more organized sharing than basic analog systems.
Services	Voice and SMS are supported; packet-data enhancements include GPRS and EDGE.
Security	Digital authentication and encryption provide better protection than 1G.

Performance measures to record: throughput (bit/s), average packet delay (ms), BER (erroneous bits / total transmitted bits), and spectral efficiency (bit/s/Hz). Use the same traffic and channel settings when comparing technologies.

6. Result Analysis

Digital coding can improve voice quality when radio errors are within the system's correction capability. Under weak signal conditions, BER rises and service quality can fall. Capacity depends on the access method, channel allocation, interference and network configuration.

7. Optimization Techniques

Use proper frequency planning, power control, channel coding, interference management and efficient handover configuration.

8. Result / Conclusion

The experiment demonstrates that 2G improved mobile communication through digital transmission, better security, increased capacity and services such as SMS, with later enhancements supporting packet data.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____